

EMBRUN, France

WMO: 075910

Lat: 44.5656N

Lon: 6.5022E

Elev: 2874

StdP: 13.23

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.0	20.3	0.3	6.2	30.2	4.3	7.6	30.7	28.2	37.4	25.0	39.2	3.6	40	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.9	88.9	63.1	86.2	62.8	83.2	62.1	65.6	83.4	64.3	81.6	63.1	79.5	10.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.8	85.3	69.6	58.3	80.8	69.1	56.9	76.9	68.5	32.0	83.7	30.9	81.5	30.0	79.7	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	20.2	17.0	12.7	92.7	4.0	2.8	9.8	94.8	7.5	96.4	5.2	98.0	2.2	100.1
			10.8	68.1	4.0	1.3	7.9	69.0	5.5	69.7	3.2	70.4	0.3	71.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	35.7	37.7	44.7	50.0	57.0	64.7	68.9	68.4	60.6	53.7	43.0	36.9	(6)
	DBStd	13.28	5.52	6.74	6.56	6.45	6.14	6.39	5.67	5.77	5.96	6.09	6.59	6.23	(7)
	HDD50	1733	444	345	187	79	12	0	0	0	3	35	221	407	(8)
	HDD65	5190	909	764	629	451	255	84	29	31	155	352	659	872	(9)
	CDD50	2412	0	1	23	78	230	441	586	571	322	148	12	0	(10)
	CDD65	395	0	0	0	0	9	75	150	137	24	0	0	0	(11)
	CDH74	4420	0	0	0	15	124	848	1657	1476	287	13	0	0	(12)
	CDH80	1388	0	0	0	1	16	243	570	515	43	0	0	0	(13)
(14) Wind	WSAvg	6.6	6.3	6.6	7.4	7.4	6.8	6.6	6.7	6.3	6.0	6.3	6.4	6.5	(14)
(15) Precipitation	PrecAvg	32.4	2.3	1.9	1.9	2.8	2.8	2.2	2.2	2.2	3.1	4.0	3.9	2.5	(15)
	PrecMax	41.4	5.0	5.3	9.2	6.7	6.8	5.9	4.8	3.8	9.8	8.7	10.2	7.1	(16)
	PrecMin	21.2	0.0	0.0	0.0	0.6	0.8	0.9	0.7	0.7	0.5	0.3	0.7	0.0	(17)
	PrecStd	5.3	1.5	1.4	1.9	1.5	1.3	1.2	1.1	0.8	2.0	2.2	2.8	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	62.6	70.0	76.0	81.7	90.1	91.5	92.5	84.5	76.0	63.9	55.5	(19)
		MCWB	41.9	45.2	49.4	54.1	60.2	64.0	63.9	64.0	61.7	56.7	49.8	42.7	(20)
	2%	DB	50.5	57.0	65.7	71.1	77.3	86.1	88.9	89.0	80.7	71.5	59.9	52.0	(21)
		MCWB	40.3	42.6	47.6	52.1	57.9	63.2	63.4	62.6	60.4	55.8	47.8	41.0	(22)
	5%	DB	47.2	52.4	61.9	67.3	73.9	82.5	86.2	85.9	77.4	68.1	56.3	48.8	(23)
		MCWB	38.4	40.6	45.9	50.3	56.4	62.1	63.1	62.3	58.7	54.4	45.9	39.7	(24)
	10%	DB	44.2	48.5	57.8	63.5	70.4	79.0	83.1	82.4	73.8	64.8	52.9	45.9	(25)
		MCWB	36.7	38.4	44.1	48.5	54.8	60.8	62.3	61.7	57.3	52.9	44.5	38.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.9	46.6	51.1	55.1	61.9	67.4	67.4	66.7	63.8	59.8	51.8	46.0	(27)
		MCDB	51.2	60.0	64.2	71.7	78.7	84.4	86.1	85.4	78.9	70.3	59.3	50.0	(28)
	2%	WB	41.8	43.8	48.8	53.1	59.4	64.8	65.7	65.1	61.8	57.7	49.4	43.5	(29)
		MCDB	48.1	54.0	62.2	68.6	74.3	81.9	83.6	83.1	77.2	67.8	56.1	48.4	(30)
	5%	WB	39.7	41.6	46.9	51.3	57.5	63.2	64.4	63.8	60.2	55.7	47.6	41.6	(31)
		MCDB	45.3	50.6	59.7	65.2	71.1	79.1	81.8	81.0	74.1	65.4	54.3	46.6	(32)
	10%	WB	37.8	39.5	45.0	49.4	55.8	61.7	63.1	62.6	58.5	54.0	45.7	39.6	(33)
		MCDB	43.0	47.2	56.0	61.6	68.0	76.5	79.8	78.4	70.9	62.3	51.7	44.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	16.2	19.1	21.3	21.3	22.0	23.9	24.9	24.8	22.9	20.0	16.8	15.4	(35)
		MCDBR	19.6	25.0	27.3	27.0	26.6	27.3	28.0	27.8	26.8	23.6	20.9	18.5	(36)
	5% WB	MCWBR	11.9	14.3	14.1	12.8	11.7	10.9	9.9	9.8	11.1	12.1	12.1	11.1	(37)
		MCWBR	17.2	22.6	24.8	25.4	24.2	25.4	25.5	24.9	24.3	21.0	18.2	15.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.258	0.283	0.328	0.370	0.383	0.388	0.377	0.367	0.352	0.331	0.293	0.260	(40)
		taud	2.408	2.371	2.339	2.284	2.308	2.330	2.361	2.395	2.422	2.473	2.497	2.448	(41)
	Ebn at Noon		279	289	288	282	281	279	281	281	276	268	262	267	(42)
		Edh at Noon	23	29	34	40	40	39	38	35	31	26	21	20	(43)
(44) All-Sky Solar Radiation	RadAvg		604	902	1247	1537	1752	2052	2124	1854	1432	946	591	514	(44)
	RadStd		79	101	159	160	149	127	142	101	115	107	93	57	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (46 neighbors)	+0.70	N/A	N/A	+1.69	N/A	N/A	N/A	N/A	N/A	+108
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page